

L^p Spaces

Functional Analysis

Morales Carmona Ronny

Autumn 2026

Definition

Suppose that $(X, \mathcal{M}, \mu)$ is a measure space, that $0 < p \leq \infty$, and that $f : X \rightarrow \mathbb{C}$ is a measurable function. Then the **p -norm**[†] of f , denoted by $\|f\|_p$, is the number defined by

$$\|f\|_p = \begin{cases} \left(\int |f|^p \right)^{1/p} & p < \infty, \\ \inf \{t \geq 0 : |f| \leq t \text{ } \mu\text{-a.e.}\} & p = \infty. \end{cases}$$

$\|f\|_\infty$ is called the **essential supremum** of f .

[†]Notice that, despite the name, $\|\cdot\|_p$ is not a norm, but a seminorm.

Example

Suppose μ is the counting measure on $\mathbb{N}$. If $z = (z_n)$ is a sequence in $\mathbb{C}$, then

$$\|z\|_p = \left(\sum_{n \in \mathbb{N}} |z_n|^p \right)^{1/p} \quad \text{and} \quad \|z\|_\infty = \sup_{n \in \mathbb{N}} \{|z_n|\}.$$

Definition

Suppose $(X, \mathcal{M}, \mu)$ is a measure space and $0 < p \leq \infty$. The **Lebesgue space** $\mathcal{L}^p(X, \mathcal{M}, \mu)$ (or simply $\mathcal{L}^p$ when the measure space is understood), is defined to be the set of measurable functions $f : X \rightarrow \mathbb{C}$ such that $\|f\|_p \leq \infty$

Example

When μ is the counting measure on $\mathbb{N}$, the set $\mathcal{L}^p$ is often denoted by ℓ^p . Thus

$$\ell^p = \{(z_n) : z_n \in \mathbb{C} \text{ and } \sum_{n=1}^{\infty} |z_n|^p < \infty\}$$

and

$$\ell^{\infty} = \{(z_n) : z_n \in \mathbb{C} \text{ and } \sup_{n \in \mathbb{N}} \{|z_n|\} < \infty\}.$$

Theorem

Suppose $(X, \mathcal{M}, \mu)$ is a measure space. Then

$$\|f + g\|_p^p \leq 2^p (\|f\|_p^p + \|g\|_p^p)$$

and

$$\|\alpha f\|_p = |\alpha| \|f\|_p$$

for all $f, g \in \mathcal{L}^p$ and all $\alpha \in \mathbb{C}$. Moreover, $\mathcal{L}^p$ is a linear space.

Definition

For $1 \leq p \leq \infty$, the **dual exponent** of p is denoted by p' and is the element of $[1, \infty]$ such that

$$\frac{1}{p} + \frac{1}{p'} = 1.$$

Theorem (Young's Inequality)

Suppose $1 < p < \infty$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

for all $a, b \geq 0$.

Theorem (Hölder's Inequality)

Suppose $(X, \mathcal{M}, \mu)$ is a measure space, $1 \leq p \leq \infty$, and $f, h : X \rightarrow \mathbb{C}$ are measurable functions. Then

$$\|fh\|_1 \leq \|f\|_p \|h\|_{p'}.$$

Theorem

Suppose $(X, \mathcal{M}, \mu)$ is a finite measure space and $0 < p < q < \infty$. Then

$$\|f\|_p \leq \mu(X)^{(q-p)/pq} \|f\|_q$$

for all $f \in \mathcal{L}^q$. Furthermore, $\mathcal{L}^q \subseteq \mathcal{L}^p$.

Theorem

Suppose $(X, \mathcal{M}, \mu)$ is a measure space, $1 \leq p < \infty$, and $f \in \mathcal{L}^p$. Then

$$\|f\|_p = \sup \left\{ \left| \int d\mu \, f h \right| : h \in \mathcal{L}^{p'} \text{ and } \|h\|_{p'} \leq 1 \right\}.$$

Theorem (Minkowski's Inequality)

Suppose $(X, \mathcal{M}, \mu)$ is a measure space, $1 \leq p \leq \infty$, and $f, h \in \mathcal{L}^p$. Then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Definition

On $\mathcal{L}^p$, let $\sim$ be the relation defined by $f \sim g$ if and only if $f = g$ μ -a.e. and thus we define

$$L^p = \mathcal{L}^p / \sim .$$

Definition

Let $\mathcal{Z}$ be the set of functions from X to $\mathbb{C}$ that equal 0 μ -a.e. and thus L^p can also be identified as the quotient space

$$\mathcal{L}^p / \mathcal{Z}.$$

Definition

For $0 < p \leq \infty$ define $\|\cdot\|_p$ on L^p by

$$\|[f]\|_p = \|f\|_p$$

for $f \in \mathcal{L}^p$.

Theorem

Suppose $1 \leq p \leq \infty$. Then $(L^p, \|\cdot\|)$ is a normed linear space.

Definition

If E is a measurable subset of X and $0 < p \leq \infty$, then we write $L^p(E)$ to denote the Lebesgue space $L^p(X, \mathcal{M}, \mu|_E)$ with $\mu|_E$ the restriction of μ to E .

Theorem

Let $1 \leq p \leq \infty$. Suppose (f_n) is a Cauchy sequence in $\mathcal{L}^p$. Then there exists $f \in \mathcal{L}^p$ such that

$$f_k \xrightarrow{L^p} f.$$

Theorem

For $1 \leq p \leq \infty$ suppose that $f \in \mathcal{L}^p$ and (f_n) is a sequence in $\mathcal{L}^p$ such that $f_k \xrightarrow{L^p} f$. Then there exists a subsequence (f_{n_k}) such that

$$f_{n_k} \xrightarrow[k \rightarrow \infty]{L^p} f$$

for almost every $x \in X$.

Theorem

Suppose $1 \leq p \leq \infty$. Then L^p is a Banach space.

Theorem

Let $1 < p \leq \infty$, let q be its conjugate, and let $h \in L^q$. Define $\varphi_h : L^p \rightarrow \mathbb{C}$ by

$$\varphi_h(f) = \int f h.$$

Then $h \mapsto \varphi_h$ is an injective linear map from L^q to $(L^p)^*$. Furthermore, $\|\varphi_h\| = \|h\|_q$ for all $h \in L^q$.

Theorem

Suppose $1 \leq p < \infty$ and let q be its conjugate. For $b = (b_n) \in \ell^q$, define $\varphi_b : \ell^p \rightarrow \mathbb{C}$ by

$$\varphi_b(a) = \sum_{k \in \mathbb{N}} a_k b_k,$$

where $a = (a_n)$. Then $b \mapsto \varphi_b$ is a bijective linear map from ℓ^q onto $(\ell^p)^*$. Furthermore, $\|\varphi_b\| = \|b\|_q$ for all $b \in \ell^q$.